

Airbnb NYC 2019 Analysis

Dataset Overview

The **Airbnb NYC 2019** dataset contains information about Airbnb listings across the five boroughs of New York City. It provides detailed information on hosts, property locations, room types, pricing, availability, and customer reviews, making it suitable for exploratory data analysis and business intelligence.

The dataset includes **48,895 Airbnb listings** with **16 attributes**, covering important aspects such as neighbourhood groups, neighbourhoods, host details, room types, listing prices, minimum stay requirements, number of reviews, reviews per month, and yearly availability.

This analysis focuses on understanding pricing patterns, neighbourhood-wise performance, room type preferences, customer review trends, host performance, and listing availability. The insights derived from the dataset help identify the most expensive neighbourhoods, the most active hosts, customer booking behaviour, and differences in listing availability across room types and locations.

Key Features Used

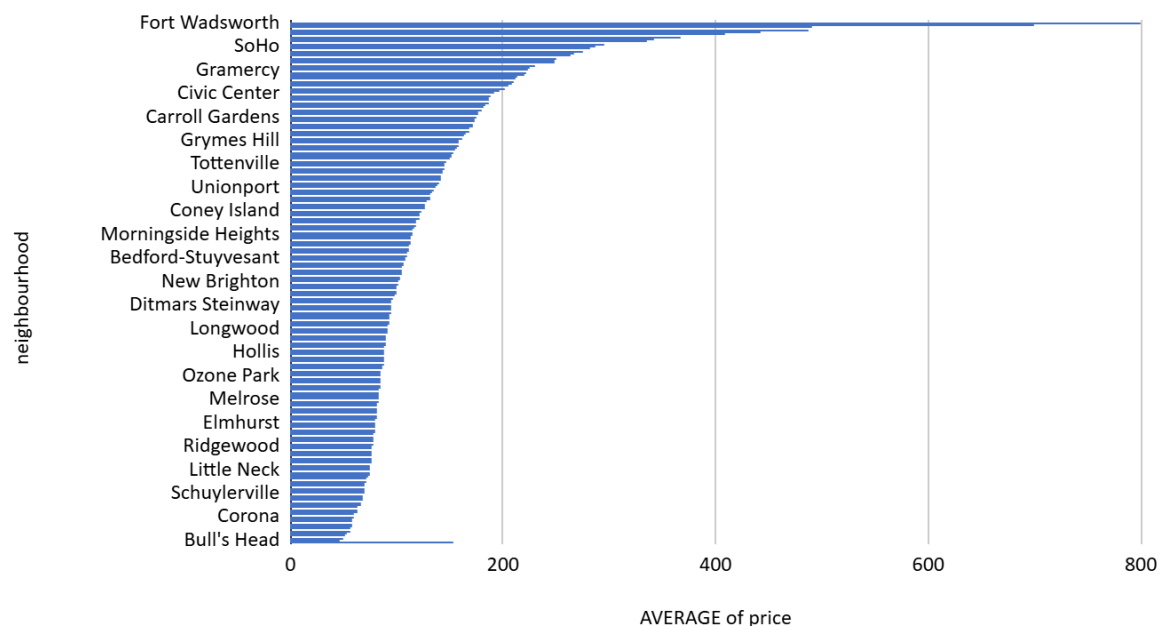
- **Neighbourhood Group** – Borough where the listing is located (Brooklyn, Manhattan, Queens, Bronx, Staten Island).
- **Neighbourhood** – Specific locality within each borough.
- **Host Name** – Name of the Airbnb host.
- **Room Type** – Type of accommodation (Entire home/apartment, Private room, Shared room).
- **Price** – Price charged per night for the listing.
- **Minimum Nights** – Minimum number of nights required for booking.
- **Number of Reviews** – Total customer reviews received.
- **Reviews per Month** – Average number of reviews received each month.
- **Availability 365** – Number of days the listing is available during the year.

Problem Statement

1. Top neighbourhoods in NYC with respect to average price/day of Airbnb listings?
2. Room types vs price in different neighbourhood groups?
3. Average minimum-night requirement by room type across neighbourhood groups?
4. How monthly reviews varies with room types in each neighbourhood groups?
5. Room -types and their relation with availability in different neighbourhood groups?
6. Top 10 reviewed hosts on the basis of reviews/month?

1. Top neighbourhoods in NYC with respect to average price/day of Airbnb listings?

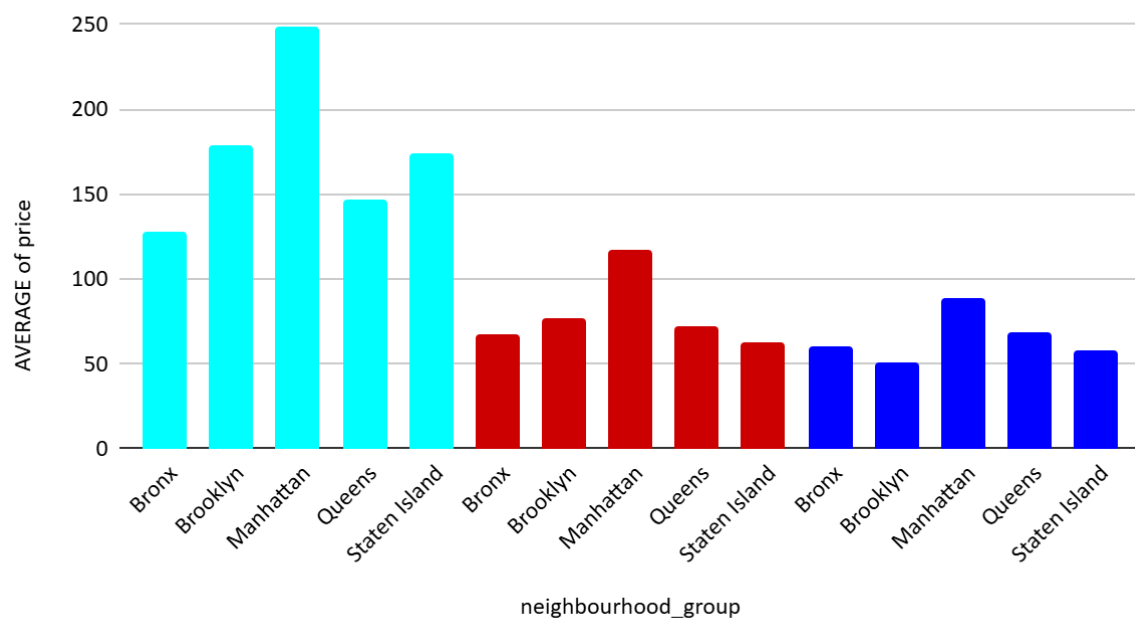
AVERAGE of price vs neighbourhood



- Premium neighbourhoods such as Fort Wadsworth and Woodrow have the highest average Airbnb prices per day.

2. Room types vs price in different neighbourhood groups?

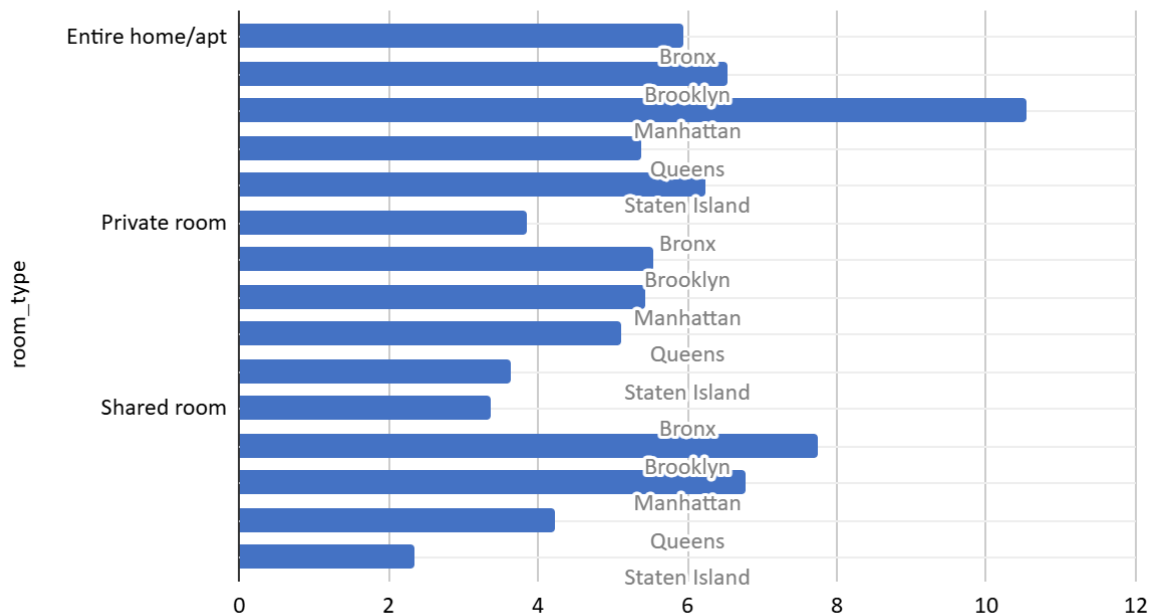
AVERAGE of price vs neighbourhood_group



- Entire homes/apartments have the highest average prices across all neighbourhood groups, with Manhattan being the most expensive.

3. Average minimum-night requirement by room type across neighbourhood groups?

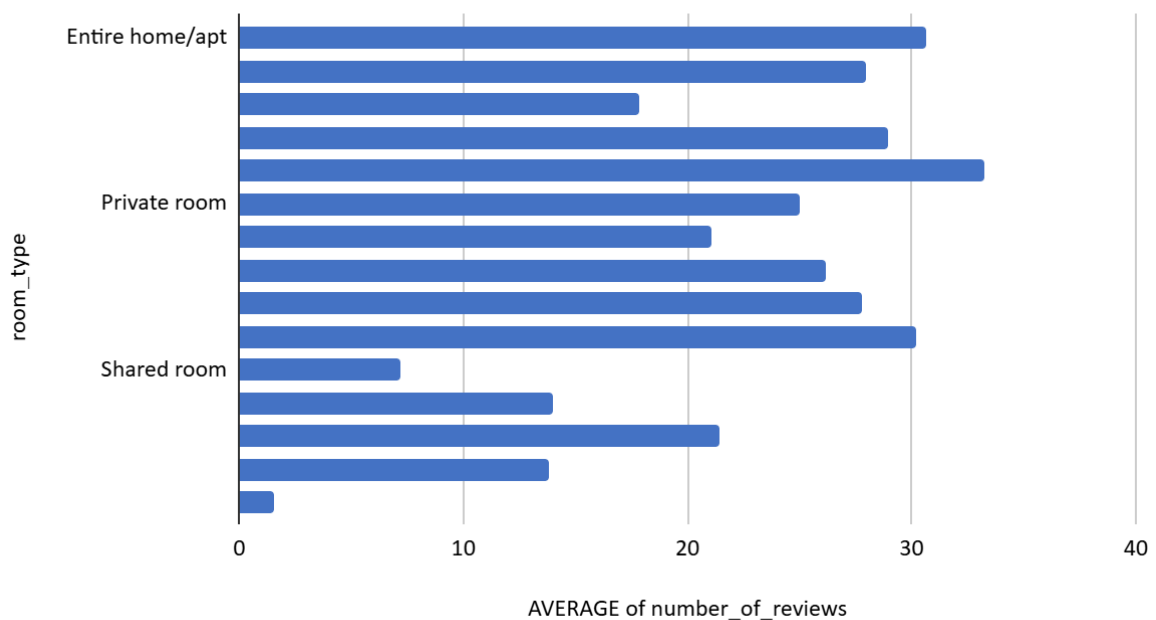
neighbourhood_group and AVERAGE of minimum_nights



- Entire homes/apartments generally require longer minimum stays than private and shared rooms, with Manhattan having the highest average minimum nights.

4. How monthly reviews varies with room types in each neighbourhood groups?

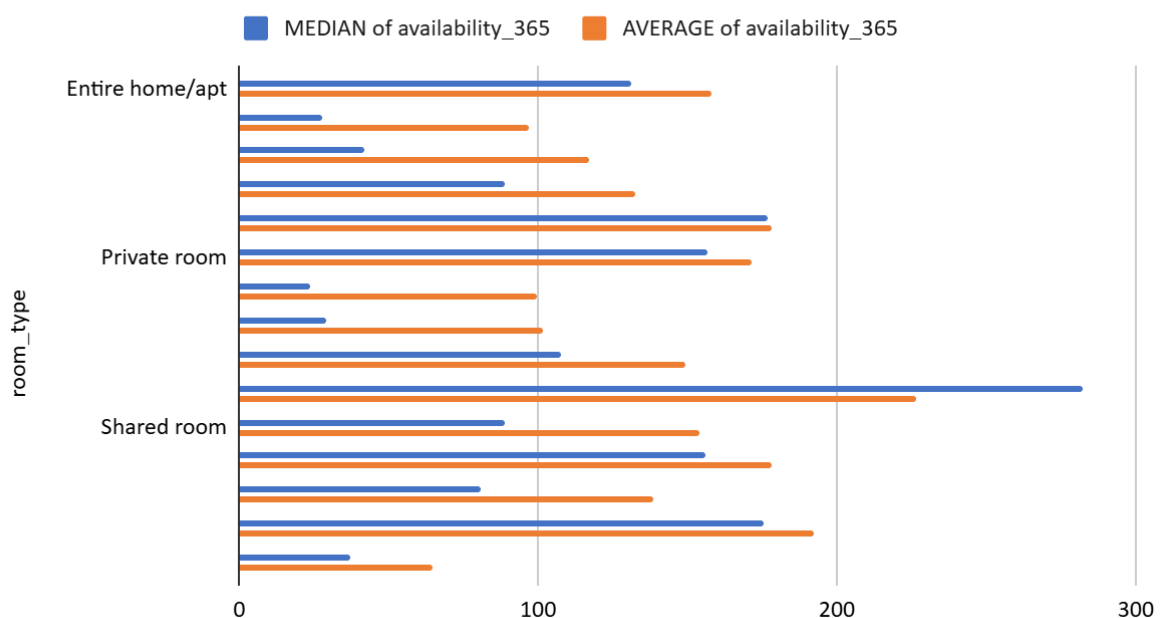
AVERAGE of number_of_reviews vs room_type



- Entire homes and private rooms generally receive more monthly reviews than shared rooms, indicating higher booking activity across most neighbourhood groups.

5. Room -types and their relation with availability in different neighbourhood groups?

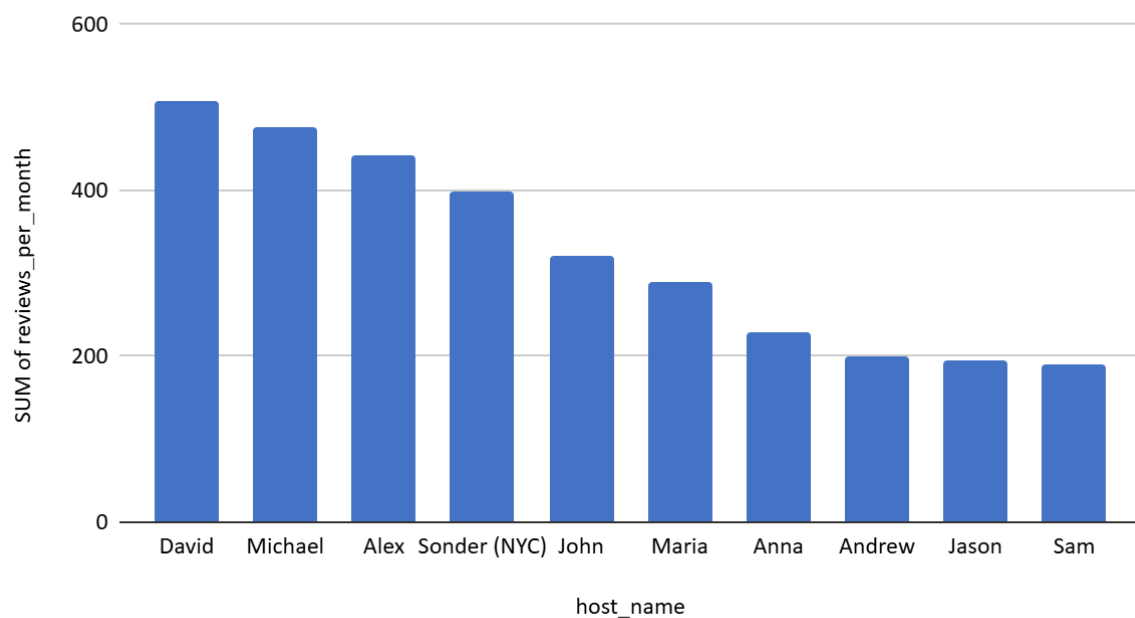
MEDIAN of availability_365 and AVERAGE of availability_365



- Listing availability varies by room type and neighbourhood group, with Shared and Private rooms generally showing higher annual availability than Entire homes in most areas.

6. Top 10 reviewed hosts on the basis of reviews/month?

SUM of reviews_per_month vs host_name



- David, Michael, and Alex receive the highest reviews per month, indicating strong guest engagement and high listing popularity.

Conclusion

- Premium neighbourhoods such as Fort Wadsworth and Woodrow have the highest average Airbnb listing prices.
- Entire homes/apartments are the most expensive room type across most neighbourhood groups, with Manhattan recording the highest average prices.
- Entire homes generally have higher minimum-night requirements than private and shared rooms.
- Entire homes and private rooms receive more monthly reviews than shared rooms, indicating higher booking activity.
- Shared and private rooms generally have higher annual availability than entire homes across most neighbourhood groups.
- David, Michael, and Alex are the top-reviewed hosts based on monthly reviews, reflecting strong guest engagement and listing popularity.